

Brief Motivation

- Quadrotors exhibit highly nonlinear, coupled dynamics making precise control inherently challenging.
- Many conventional control strategies struggle to achieve accurate and stable trajectory tracking under real-world disturbances and uncertainties.
- There is a need for advanced control system design to improve performance, stability, and responsiveness.
- Addressing this challenge is essential for enabling reliable deployment of quadrotors in applications such as surveillance, mapping, and aerial robotics.

Objective

- To design a control system for quadrotors that ensures stable flight under nonlinear dynamics.
- To develop control strategies for accurate trajectory tracking.
- To implement the designed controller in a simulation environment.
- To analyze the performance of the controller under disturbances and uncertainties.
- To evaluate tracking accuracy, stability, and response characteristics.

Methodology

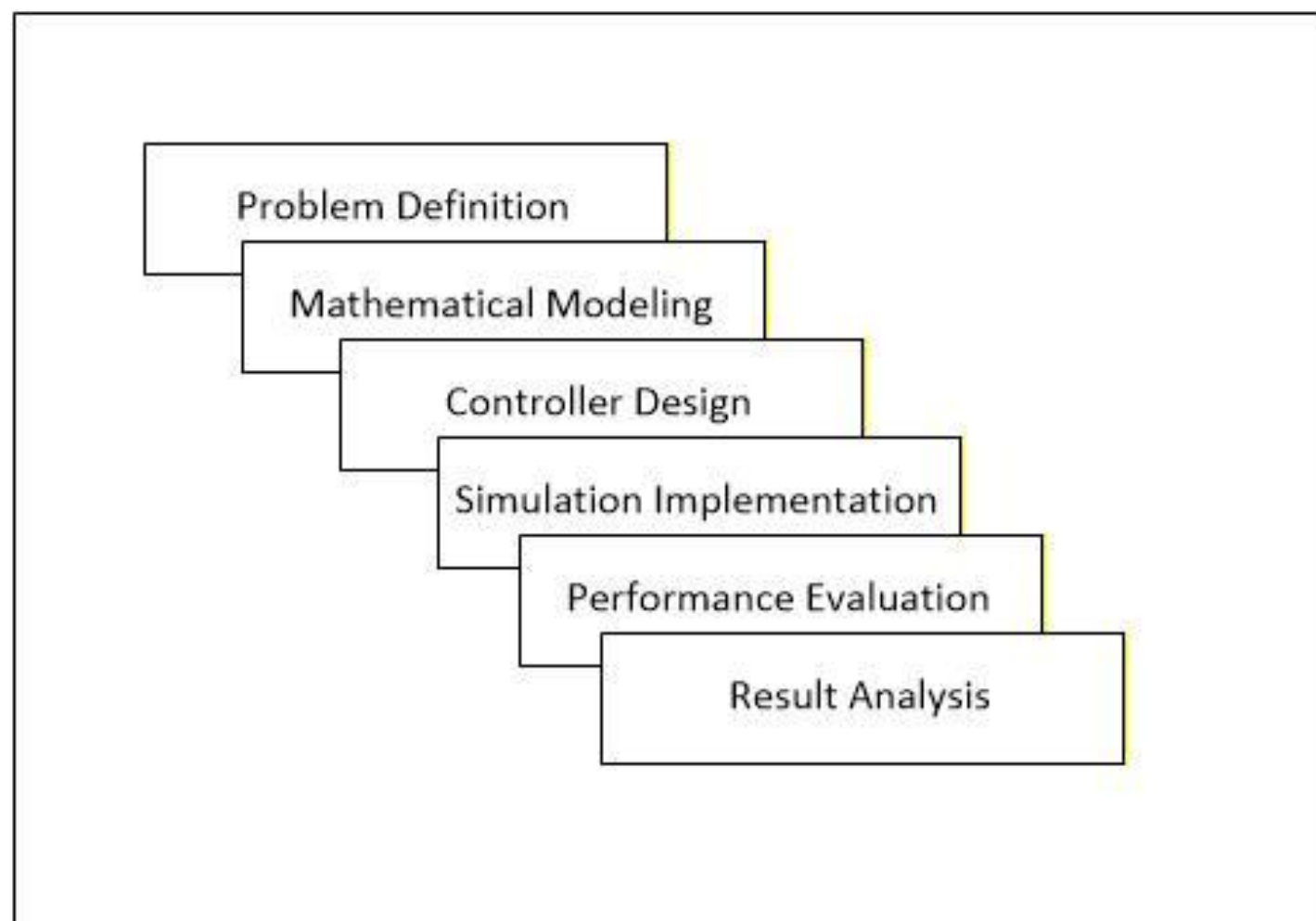


Fig.: Workflow

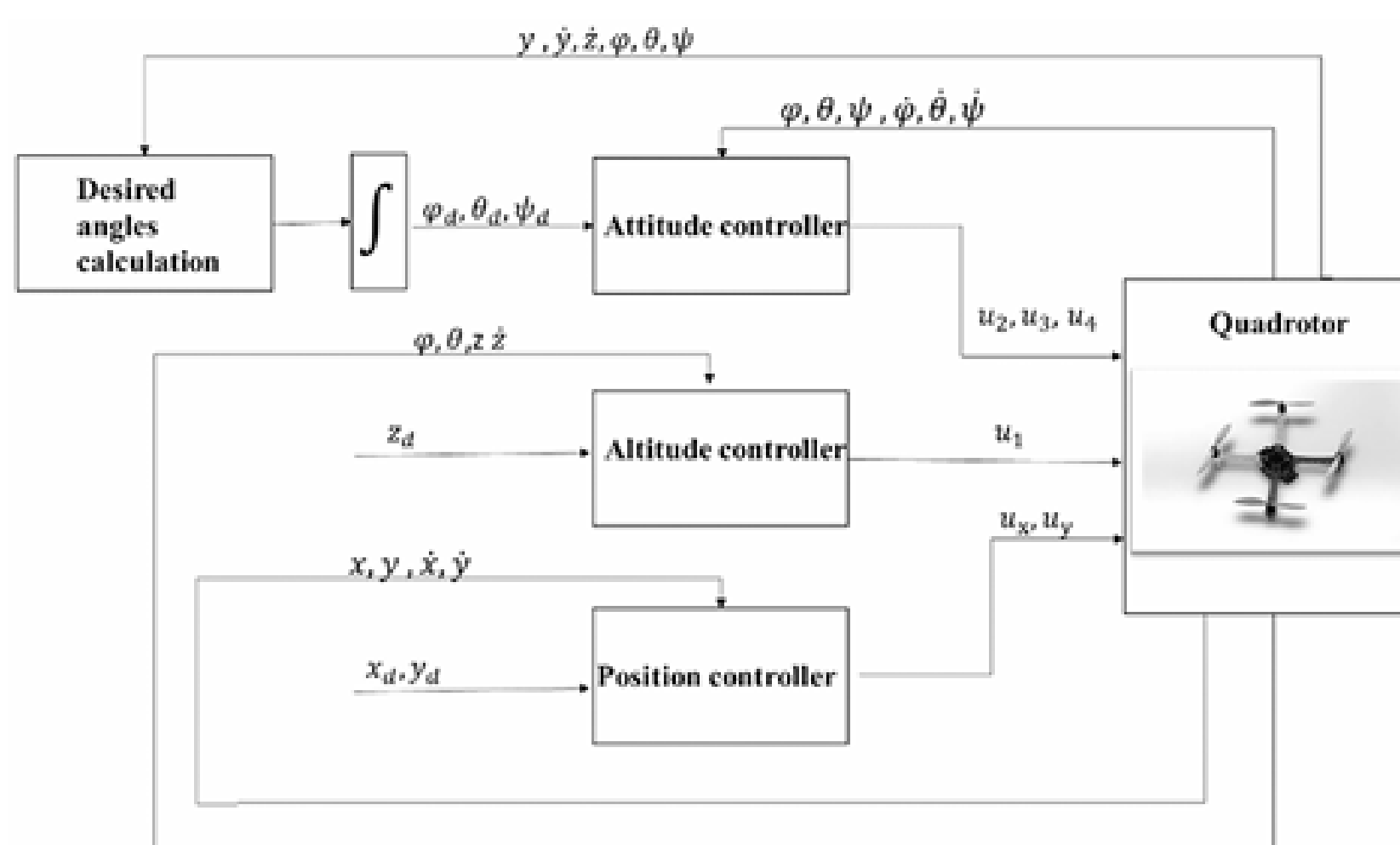


Fig.: Control System Design Block

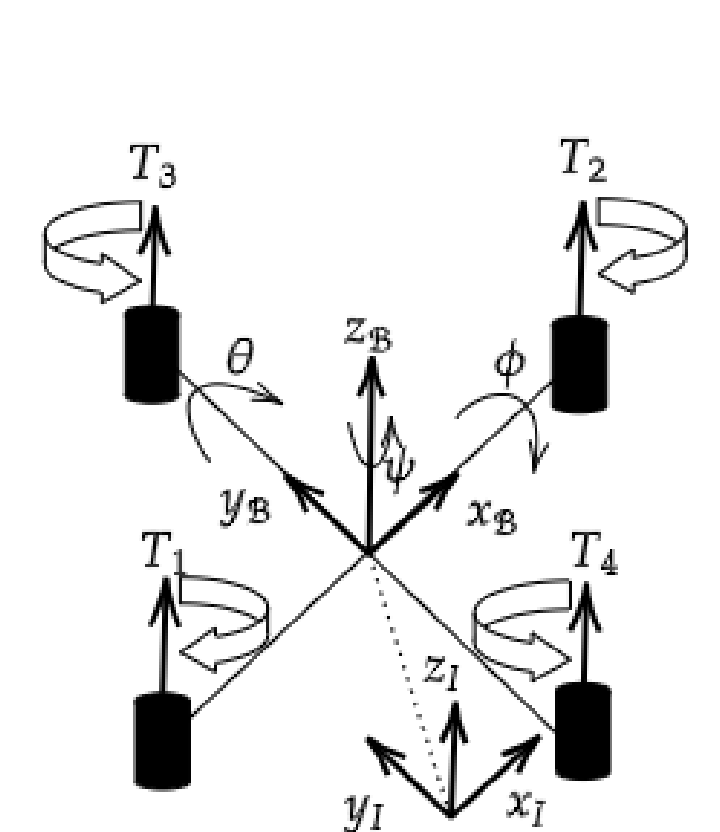


Fig.: Quadrotor configuration

Translational and Rotational Dynamics

$$m\ddot{\mathbf{p}} = R(\phi, \theta, \psi)[0 \ 0 \ T]^T - [0 \ 0 \ mg]^T$$

$$I\dot{\omega} + \omega \times (I\omega) = \tau$$

Super Twisting Sliding Mode Controller

$$s = e + \lambda e \quad u = -k_1 |s|^{\frac{1}{2}} \text{sign}(s) + v$$

$$u = -k \text{sign}(s) \quad v = -k_2 \text{sign}(s)$$

Expected Results

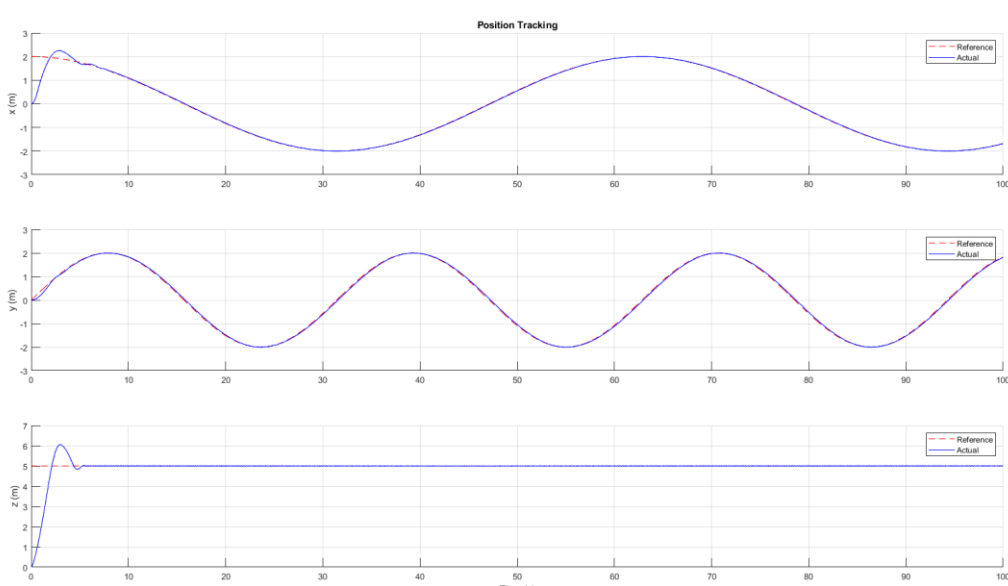


Fig.: Position Tracking

Fig.: Attitude Tracking

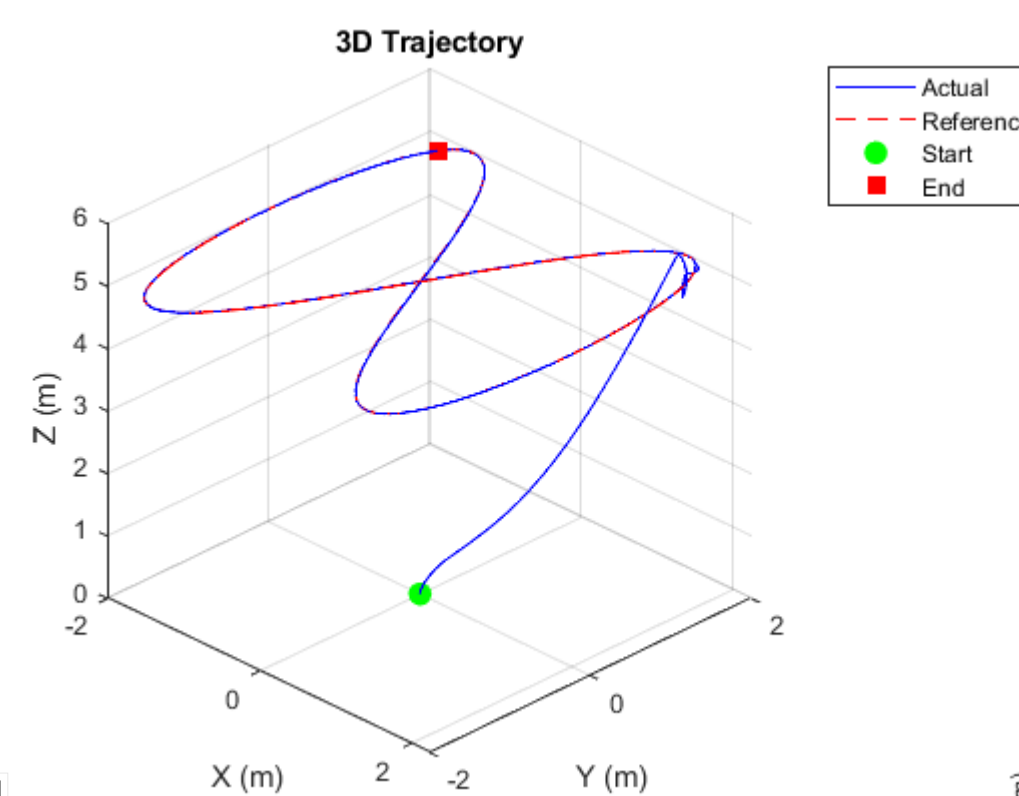
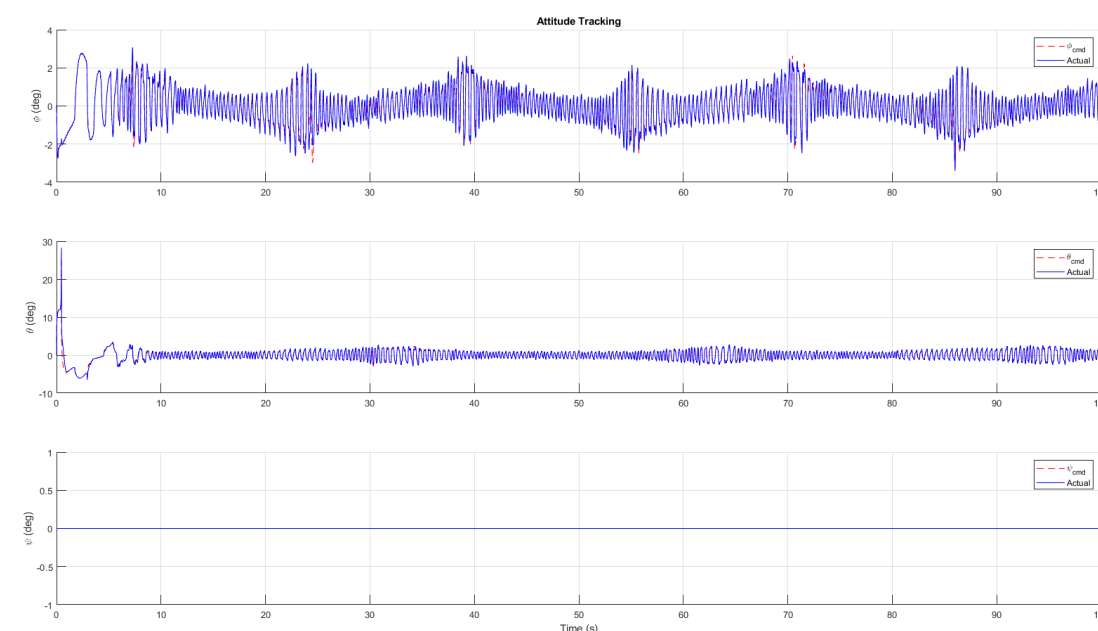
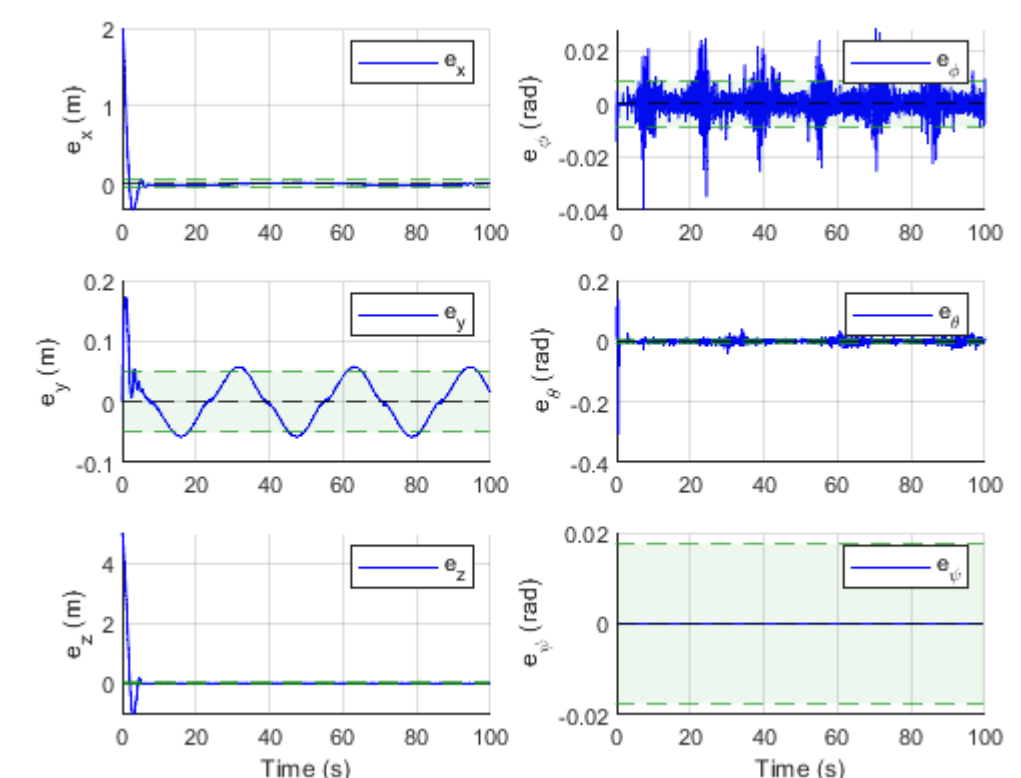


Fig.:3D Trajectory

Fig.: Error Plots



Outcomes

- Understand nonlinear dynamics and control challenges associated with quadrotor systems.
- Apply control strategies (SMC and ST -SMC) for trajectory tracking and stabilization.
- Implement quadrotor models and controllers in simulation using MATLAB.
- Evaluate the effectiveness of different control approaches through simulation results.

References

1. Kumar, S., Yesmin, A., Tripathy, T., Sinha, A., & Guha, A. (2025, December). Disturbance Analysis with Super-Twisting Sliding Mode Control for UAVs. In *2025 Eleventh Indian Control Conference (ICC)* (pp. 660-665). IEEE.
2. Yesmin, A., & Sinha, A. (2025). Sliding Mode Controller for Quadcopter UAVs: A Comprehensive Survey. *Drones*, 9(9), 625. <https://doi.org/10.3390/drones9090625>.
3. Yesmin, A., Kumari, K., Borkar, A. V., Sinha, A., & Arya, H. (2022). Software-In-The-Loop validation of super twisting based sliding mode control for quadcopters. *IFAC-PapersOnLine*, 55(22), 123-128.
4. Olguin-Roque, J., Salazar, S., González-Hernandez, I., & Lozano, R. (2023). A robust fixed-time sliding mode control for quadrotor UAV. *Algorithms*, 16(5), 229.